

PART-A

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Signature of the Student: 

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Signature of the Invigilator: 

INSTRUCTIONS

1. Check that you have **10 printed pages**. The PART-A of the examination has **3 questions**, for a total of **20 points**.
2. Attempt **all questions**.
3. There is **no credit** for a solution if the appropriate work is not shown, even if the answer is correct.
4. Write your answers in this booklet and only in the **space marked for answers**. Any answer written in any space other than the designated space will **not be evaluated**.
5. The pages numbered **8 to 10** are kept as **additional pages**. If you cannot able to fit a complete answer in the space provided just after the question, you can use these three pages by referring the page number of additional page in the original space.
6. At the beginning of the examination, a supplementary sheet has been provided only for rough work. Should you need more, please ask the invigilator.
7. No question requires any clarification from the instructors or invigilators, even if a question has an error or incomplete data. The students are advised to write answer according to their understanding or write reasons for why it is not possible to solve partially or completely that question by citing errors/insufficient data.

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Question:	1	2	3	Total
Points:	5	6	9	20
Score:	5	3+1	8	17

QUESTIONS

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$Y_1 = \theta_1 + \theta_2 + \epsilon_1,$$

$$Y_2 = \theta_1 - 2\theta_2 + \epsilon_2,$$

$$Y_3 = 2\theta_1 - \theta_2 + \epsilon_3,$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Now $X = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}$ (Given)

Let $\hat{\theta} = \begin{bmatrix} \hat{\theta}_1 \\ \hat{\theta}_2 \end{bmatrix}$

$$X^T = \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$\therefore X^T X = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}$$

$$\therefore (X^T X)^{-1} = \begin{bmatrix} \frac{2}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{2}{9} \end{bmatrix}$$

$$\therefore (X^T X)^{-1} X^T = \begin{bmatrix} \frac{2}{9} & \frac{1}{9} \\ \frac{1}{9} & \frac{2}{9} \end{bmatrix} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix}$$

Now $\hat{\theta} = (X^T X)^{-1} X^T Y$

$$\therefore \hat{\theta} = \begin{bmatrix} \frac{1}{3} & 0 & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} \frac{y_1 + y_3}{3} \\ \frac{y_1 - y_2}{3} \end{bmatrix}$$

$$\therefore \hat{\theta}_1 = \frac{y_1 + y_3}{3} \text{ is the LSE of } \theta_1$$

$$\hat{\theta}_2 = \frac{y_1 - y_2}{3} \text{ is the LSE of } \theta_2$$

05

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

(a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

(b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

$$\begin{aligned} \text{q) Var}(X_1) &= E[X_1^2] - E^2[X_1] \quad \because X_1 = U_1 \\ &= \int_0^1 x^2 dx - \left[\int_0^1 x dx \right]^2 \\ &= \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{3} - \frac{1}{4} = \frac{1}{12} \end{aligned}$$

$$\text{Similarly Var}(X_2) = \frac{1}{12}$$

$$\begin{aligned} \text{Var}(X_3) &= \text{Var}(U_1 + U_2) = \text{Var}(U_1) + \text{Var}(U_2) + 2\text{Cov}(U_1, U_2) \\ &= \frac{1}{12} + \frac{1}{12} + 0 \quad \because U_1 \text{ and } U_2 \text{ are independent} \\ &= \frac{1}{6} \end{aligned}$$

$$\text{Now Cov}(X_1, X_2) = \text{Cov}(U_1, U_2) = 0$$

$$\begin{aligned} \text{Cov}(X_1, X_3) &= \text{Cov}(U_1, U_1 + U_2) = \text{Cov}(U_1, U_1) + \text{Cov}(U_1, U_2) \\ &= \text{Var}(U_1) + 0 \\ &= \frac{1}{12} \end{aligned}$$

$$\text{Cov}(X_2, X_3) \text{ similarly} = \frac{1}{12}$$

$\therefore$ The variance covariance matrix is

$$\begin{bmatrix} \frac{1}{12} & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} \end{bmatrix}$$

$$\text{Now } \rho_{11} = \rho_{22} = \rho_{33} = 1$$

$$\rho_{12} = \frac{\text{Cov}(X_1, X_2)}{\sigma_1 \sigma_2}$$

$$= 0$$

$$\rho_{13} = \frac{\text{Cov}(X_1, X_3)}{\sigma_1 \sigma_3}$$

$$= \frac{\frac{1}{12}}{\frac{1}{\sqrt{12}} \cdot \frac{1}{\sqrt{6}}} = \frac{\frac{\sqrt{12}}{12}}{\frac{1}{\sqrt{12}} \cdot \frac{1}{\sqrt{6}}} = \frac{1}{\sqrt{2}}$$

$$\rho_{23} = \frac{1}{\sqrt{2}}$$

$$\therefore R = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}$$

b) Solving characteristic eqn $|R - \lambda I| = 0$ gives us λ 's eigenvalues 2, 1, 0 * (last page (additional page no.)) How?

$\therefore$ Solving eigen vector corresponding to $\lambda = 2$

$$Rv = \lambda v \Rightarrow \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} 2v_1 \\ 2v_2 \\ 2v_3 \end{bmatrix}$$

$$\therefore v_1 + \frac{v_3}{\sqrt{2}} = 2v_1 \Rightarrow v_3 = \sqrt{2}v_1 \quad (1)$$

$$v_2 + \frac{v_3}{\sqrt{2}} = 2v_2 \Rightarrow v_3 = \sqrt{2}v_2 \quad (2) \Rightarrow v_1 = v_2 = \frac{v_3}{\sqrt{2}} \quad (3)$$

$$\frac{v_1 + v_2}{\sqrt{2}} + v_3 = 2v_3 \Rightarrow v_3 \sqrt{2} = \frac{v_1 + v_2}{\sqrt{2}} \quad (3) \text{ derived from } (1) \text{ and } (2) \text{ only}$$

$$(4) \text{ Using } v_1^2 + v_2^2 + v_3^2 = 1$$

$$\Rightarrow v_3^2 + \frac{v_3^2}{2} + \frac{v_3^2}{2} = 1 \Rightarrow v_3^2 = \frac{1}{2} \Rightarrow v_3 = \frac{1}{\sqrt{2}}, v_1 = \frac{1}{2}, v_2 = \frac{1}{2}$$

$$\therefore e'_1 = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$\text{Now } u_1 = e'_1 x$$

$$\therefore \text{Var}(u_1) = \text{Var}(e'_1 x) = e'_1 \text{Var}(x) e_1$$

$$= \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \end{bmatrix} \begin{bmatrix} \frac{1}{12} & 0 & \frac{1}{12} \\ 0 & \frac{1}{12} & \frac{1}{12} \\ \frac{1}{12} & \frac{1}{12} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{\sqrt{2}+1}{24} & \frac{\sqrt{2}+1}{24} & \frac{\sqrt{2}+1}{12} \end{bmatrix} \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$= \left[\frac{(\sqrt{2}+1)}{48} + \frac{(\sqrt{2}+1)}{48} + \frac{2+\sqrt{2}}{24} \right] = \frac{4+2\sqrt{2}+\sqrt{2}+\sqrt{2}+2}{48} = \frac{6+4\sqrt{2}}{48} = \boxed{6.24285}$$

3. Let the matrix of distance be

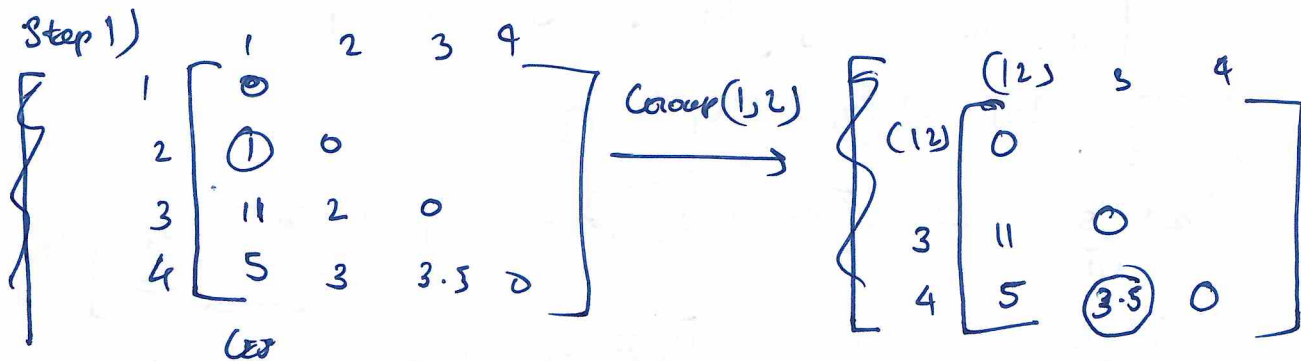
	1	2	3	4
1	0			
2	1	0		
3	11	2	0	
4	5	3	3.5	0

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

(b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Q) i) Complete Linkage $d(u,v,w) = \max(d_{uw}, d_{vw})$

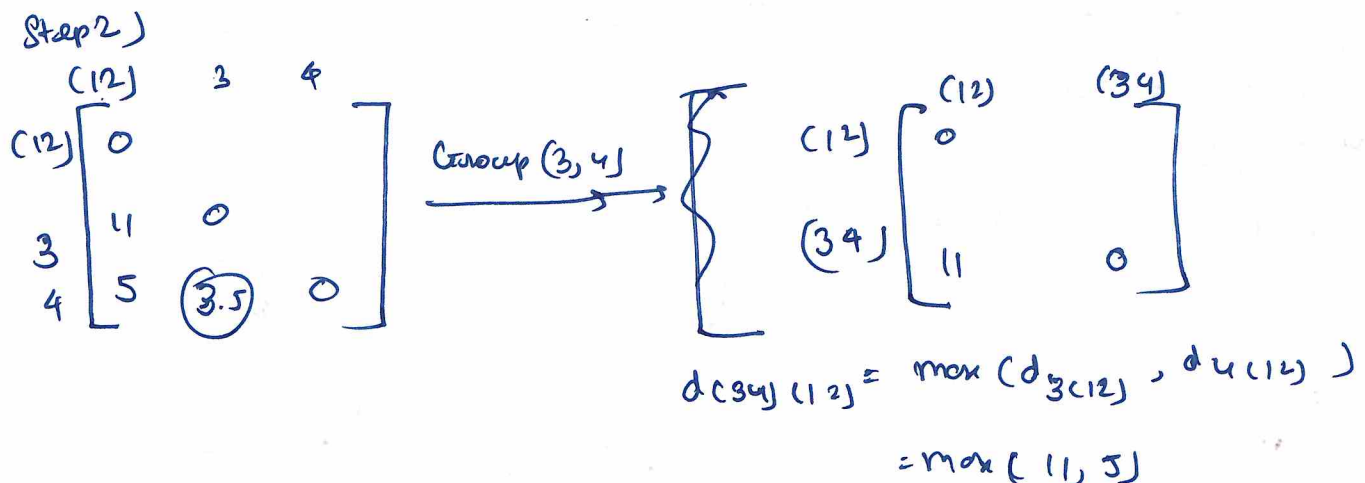


$$d_{(12)3} = \max(d_{13}, d_{23})$$

$$= \max(11, 2) = 11$$

$$d_{(12)4} = \max(d_{14}, d_{24})$$

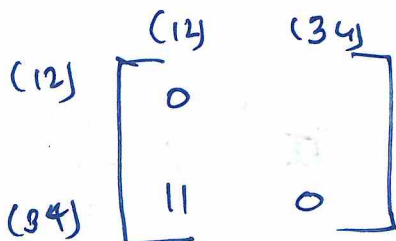
$$= \max(5, 3) = 5$$



$$d_{(123)(34)} = \max(d_{3(12)}, d_{4(12)})$$

$$= \max(11, 5)$$

Thus the final clustering is (Using Complete Linkage) $= \max 11$



ii) Average Linkage $d_{uv} = \frac{\sum_i \sum_j d_{ij}}{N_{uv} \cdot N_w}$

Step 1

$$\begin{matrix} & \begin{matrix} 1 & 2 & 3 & 4 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & & \\ \textcircled{1} & 0 & & \\ 11 & 2 & 0 & \\ 5 & 3 & 3.5 & 0 \end{bmatrix} \end{matrix} \xrightarrow{\text{Group } (1,2)}$$

$$N_{12} = N_{12} = 2$$

$$N_{12} = 1$$

$$d_{(12)3} = \frac{11 + 2}{2 \cdot 1} = 6.5$$

$$d_{(12)4} = \frac{5 + 3}{2 \cdot 1} = 4 \quad \underline{N_{(4)} = 1}$$

$$(12) \begin{matrix} & \begin{matrix} 3 & 4 \end{matrix} \\ \begin{matrix} 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & \\ 6.5 & 0 \\ 4 & \textcircled{3.5} \end{bmatrix} \end{matrix}$$

Step 2

$$\begin{matrix} & \begin{matrix} (12) & 3 & 4 \end{matrix} \\ \begin{matrix} (12) \\ 3 \\ 4 \end{matrix} & \begin{bmatrix} 0 & & \\ 6.5 & 0 & \\ 4 & \textcircled{3.5} & 0 \end{bmatrix} \end{matrix} \xrightarrow{\text{Group } 3,4}$$

$$d_{(34)(12)} = \frac{6.5 + 4}{2 \cdot 2} \quad \therefore N_{(34)} = 2$$

$$N_{(12)} = 2$$

$$= \frac{6.5 + 4}{2} = 5.25$$

06

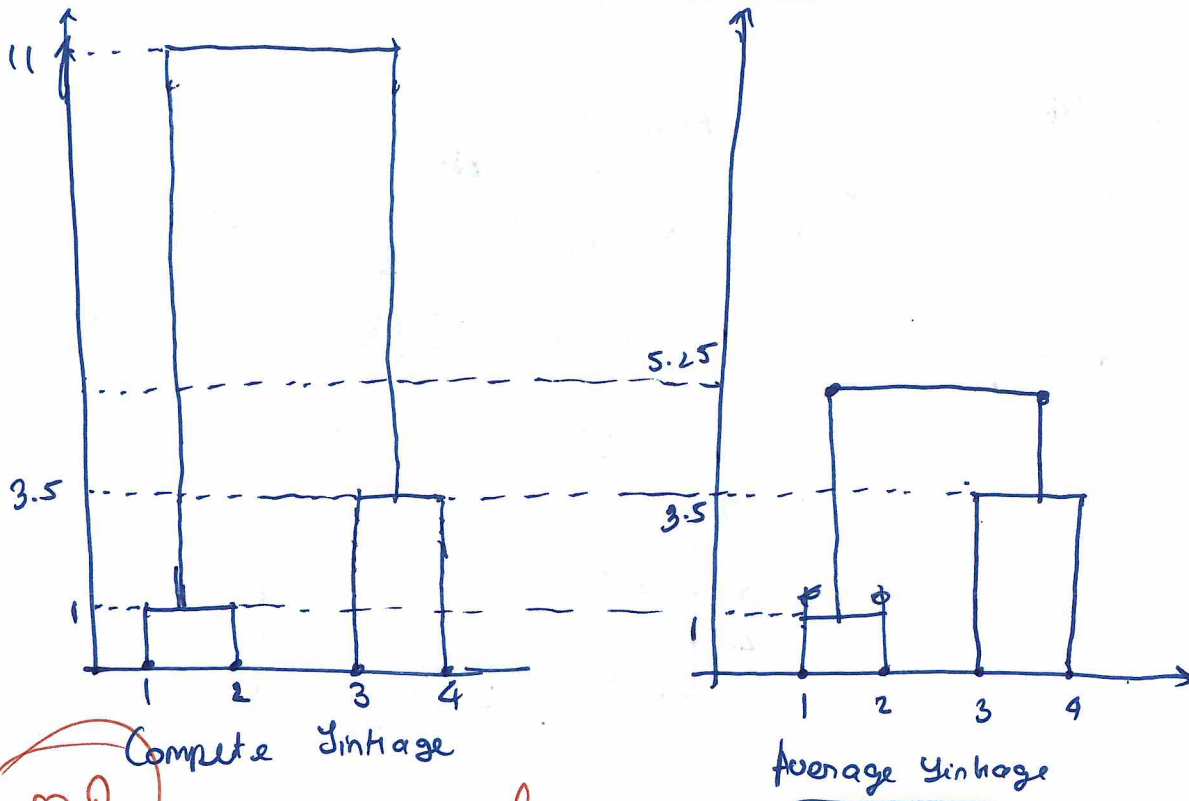
$$\begin{matrix} & \begin{matrix} (12) & (34) \end{matrix} \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{matrix}$$

Final clustering using Complete Average Linkage

$$\begin{matrix} & \begin{matrix} (12) & (34) \end{matrix} \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{matrix}$$

Can Contd. on next page

Q.3) b) Pen do grow for Complete Linkage and Average Linkage



02

Comments ??

Q.2) b) (*) This is for calculation of ~~showing~~ the eigenvalues

$$|R - \lambda I| = 0 \Rightarrow \begin{vmatrix} 1-\lambda & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} = 0$$

$$\Rightarrow (1-\lambda) \left((1-\lambda)^2 - \frac{1}{2} \right) + \frac{1}{\sqrt{2}} \left(\frac{1-\lambda}{\sqrt{2}} \right) = 0$$

$$\Rightarrow (1-\lambda)^3 - \frac{(1-\lambda)}{2} - \frac{(1-\lambda)}{2} = 0$$

$$\Rightarrow (1-\lambda)^3 - (1-\lambda) = 0$$

$$\Rightarrow 1 - \lambda^3 + 3\lambda^2 - 3\lambda + 1 = 0$$

$$\Rightarrow \lambda^3 - 3\lambda^2 + 2\lambda = 0 \Rightarrow \lambda(\lambda-1)(\lambda-2) = 0$$

$$\Rightarrow \lambda = 0, 1, 2$$

01

77

